

Unit 7, Lesson 8: Simple Interest

Benchmark

MA.7.AR.3.1 Apply previous understanding of percentages and ratios to solve multi-step real-world percent problems.

Learning Targets

- ☐ I can solve problems that involve simple interest.
- ☐ I can compare simple interest and compound interest.

7.8.1 Warm-Up: Math Fail

The cost for dinner at a restaurant was \$23.60. As a courtesy, the restaurant listed suggested tips for customers on the receipt. The receipt stated a 15% tip was \$6.53, an 18% tip was \$7.83, and a 20% tip was \$8.70. The customer underlined the recommended 15% tip amount and wrote, "No it's not."



1. Explain whether the restaurant or customer is correct. Justify your answer with calculations.

7.8.2 Guided Instruction: Simple Interest

1. Kristina received \$100 in a birthday card and wants to put the money into her savings account, where she will earn an additional 2% each year it remains in her account. This is known as ***simple interest***.



Simple interest is a method of computing interest. Interest is computed from the (original) principal alone, no matter how much money has accrued so far.

- a. How much is 2% of \$100?
 - b. How much money will be in Kristina's savings account after one year if she makes no additional deposits or withdrawals after putting in her birthday money?
 - c. Compare your answer with your partner and discuss how you determined the answer for Part B.
2. Suppose an investment of \$100 earns \$3.50 interest in one year.
 - a. What investment amount will earn \$61.60 in two years at the same rate?
 - b. Ask two other partner pairs to describe how they found the initial investment. Record their initial investment and write your summary of their process. *You are the only person who should write in the first two columns of the table.* Have the person initial next to your summary stating that your summary was correct.

Initial Investment	My Summary of Their Reason	Initials

- c. Review the initial investments and reasons provided by other partner pairs. If necessary, fix any mistakes you and your partner made.



Simple Interest Formula

$$I = p \times r \times t$$

where I is the interest earned, p is the principal starting amount of money, r is the interest rate per year, and t is the length of the investment.

3. Tonio took out a six-year car loan for \$15,600, with a 3% simple interest rate over the life of the loan.
- a. How much interest, in dollars, will Tonio pay annually?
- b. How much will Tonio pay in interest over the course of **six** years?

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- c. What is the total amount Tonio will pay back at the end of his six-year loan?
- d. Explain why the total amount Tonio will have paid back by the end of his loan is more than the loan amount he borrowed.

7.8.3 Try It: Simple Interest

1. Quana is researching rates for investing \$3,500 over a 3-year period.

How much money would Quana earn on an investment that offers a 4% simple interest rate?

2. Sanjay has \$750 he wants to put into a savings account.

How much would he have at the end of 5 years in an account that offers a 2% simple interest rate?

7.8.4 Collaboration: Comparing Interest

Not all interest is simple interest; there is also **compound interest**. Compound interest is interest earned on interest over time.

With your partner, complete the following.

1. Charlie wants to take out a \$5,750 loan for new furniture. The bank offers him two different options for his loan. He can take out a loan for three years at a rate of 4% compounded annually, or he can take out a loan for four years at 5% simple interest.

When a loan is compounded, that means interest owed each year is based on the principal *and* any accrued interest, not the principal value alone.

- a. The compound interest table has been filled out. Complete the simple interest table to help Charlie choose the loan he will pay the least.

4% Compound Interest	
Year 1	$5750 \cdot 1.04 = 5,980$
Year 2	$5750 \cdot 1.04 \cdot 1.04 = 6219.20$
Year 3	$5750 \cdot 1.04 \cdot 1.04 \cdot 1.04 = 6,467.97$

5% Simple Interest	
Year 1	$5750 \cdot 0.05 \cdot 1 = 287.50$ $287.50 + 5750 = 6,037.50$
Year 2	$5750 \cdot 0.05 \cdot 2 =$
Year 3	
Year 4	

- b. Explain which loan Charlie should choose.

2. The two tables summarize different options Valentina has to invest \$1,000. Use the tables to answer the questions.

Simple Interest		
Years Invested	Calculation	Yearly Total (rounded to nearest whole)
1	$\$1000 + (\$1,000 \cdot 0.07 \cdot 1)$	\$1,070
2	$\$1000 + (\$1,000 \cdot 0.07 \cdot 2)$	\$1,140
3	$\$1000 + (\$1,000 \cdot 0.07 \cdot 3)$	\$1,210
4	$\$1000 + (\$1,000 \cdot 0.07 \cdot 4)$	\$1,280
5	$\$1000 + (\$1,000 \cdot 0.07 \cdot 5)$	\$1,350

Compound Interest		
Year	Calculation	Yearly Total (rounded to the nearest whole)
1	$\$1,000 (1.07)$	\$1,070
2	$\$1,000 (1.07) (1.07)$	\$1,145
3	$\$1,000 (1.07) (1.07) (1.07)$	\$1,225
4	$\$1,000 (1.07) (1.07) (1.07) (1.07)$	\$1,311
5	$\$1,000 (1.07) (1.07) (1.07) (1.07) (1.07)$	\$1,403

- a. How are the two tables similar?
- b. How are they different?
- c. How are these similarities and differences represented in the calculation column for each scenario?

Find another partner pair and discuss the following.

- d. How does the growth in an account that calculates compound interest compare to the growth in an account that calculates simple interest?

7.8.5 Wrap-Up: Cool Down

1. Your friend is absent and will need to make up the lesson from today. Write a quick note to help them understand what simple interest is, how to calculate it, and how to use it to solve problems.